

Part H

Landfill Guidelines

Blacktown City Council

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1 Introduction

Activities associated with illegal landfilling are a common form of unauthorised activity within Blacktown City. Such incidents often occur as landowners and/or occupants are uncertain of activities that are allowed on land, and this can be compounded if construction and demolition industry contractors seek to avoid associated waste disposal costs. Illegal landfill activity also occurs as the result of owners seeking to develop on low-lying property, often flood affected or traversed by a stormwater drainage system or watercourse.

Activities associated with illegal landfilling/excavation can have adverse effects, such as changes to existing overland flow patterns, vegetation loss, increased flood levels, land instability and sedimentation. These can negatively affect upstream and downstream landowners. Council has sought to address these issues through the controls contained within this Part of the DCP.

Specifically, as a key requirement of this section, Council requests that the purpose of any related landfill activity is clearly stated on the Development Application (DA) and accompanying documentation. Furthermore, Council has a clearly stated requirement for a neutral net/cut landfill approach, the details of which are outlined within this section.

1.1 Scope

These landfill guidelines identify the requirements for undertaking activities associated with landfill on rural lands within the Blacktown local government area.

1.2 Aims

The aims of this Part of the DCP are to:

(a) Provide guidelines with respect to proposed landfill activities

(b) Promote acceptable bulk earthworks practices for building site preparation

(c) Control and minimise variations to natural ground levels

(d) Ensure that any landfill-related activity does not independently or cumulatively adversely affect the environment.

1.3 Objectives

The main objectives of this Part are to ensure that all landfill activity:

(a) Enhances a development and/or use of land

(b) Does not adversely affect local stormwater drainage or change flood prone land characteristics

(c) Does not contaminate land

(d) Utilises clean material where landfilling is required

(e) Is suitably treated to minimise soil erosion and weed infestation

(f) Provides for the installation of sedimentation controls to minimise the potential for water pollution

(g) Does not injure or destroy any tree

(h) Considers the visual impact on a locality by complementing adjoining land features

(i) Does not adversely affect the health and safety of any person

(j) Considers the relevant State and Commonwealth Government legislative requirements, appropriate guidelines and Council policies

(k) Does not impact upon threatened species and ecological communities protected under the *Biodiversity Conservation Act 2016*.

1.4 Definitions

Definitions or terms used within this Part are provided below.

Annual exceedance probability (AEP) flood means the chance of a flood of a given or larger size occurring in any one year, usually expressed as a percentage. For example, a 1% AEP flood has a 1% or 1 in 100 chance of being reached or exceeded in any given year Annual recurrence interval (ARI) is the historical way of describing a flood event.

Activity means any development on land and may include any one of the following:

(a) The erection of a building

(b) The carrying out of a work in, on, over or under land

(c) The use of land or of a building or work

(d) The subdivision of land involving earthworks

(e) Any soil disturbing activity in or on a public place or on land owned by Council which may or may not be the subject of an approval

(f) Any act, matter or thing for which provision may be made under Section 26 of the EP&A Act 1979 and which is prescribed for the purpose of this definition, but does not include:
  (i) Any act, matter or thing referred to in Section 26 of the EP&A Act for which development consent is required or has been obtained
  (ii) Any act, matter or thing that is prohibited under an environmental planning instrument.

Approval / approved means a consent, license or permission or any form of authorisation issued by Blacktown City Council.

Biodiversity is defined as the variety of living animal and plant life from all sources. It includes the diversity of ecosystems and the diversity of species supported within them.

Biodiversity Development Assessment Report (BDAR) is a report prepared by an Accredited Biodiversity Assessor consultant. The report identifies how a proponent of a Development Application (DA) or State Significant Development (SSD) in NSW, will avoid, minimise and offset impacts to native vegetation or threatened species from their proposed development.

Biodiversity Management Plan is a plan of actions and mitigation measures to be undertaken to reduce the impact on biodiversity prior to, during and following land clearing for remediation, subdivision or construction works.

Biodiversity Offset Scheme (BOS) a statewide scheme which provides a mechanism to avoid, minimise and offset the impacts of development and some types of clearing on biodiversity in New South Wales.

Biodiversity Values Map (BVM) identifies land with high biodiversity value, such as native vegetation, threatened species habitat and creek lines, that is particularly sensitive to impacts from development. Landfill works in these areas will trigger entry to the Biodiversity Offsets Scheme.

Clean landfill means **virgin excavated natural material (VENM)**. Refer to separate definition below.

Contamination / contaminated landfill means the presence in, on or under the land of a substance at a concentration above the concentration at which the substance is normally present in, on or under (respectively) land in the same locality, being a presence that presents a risk of harm to human health or any other aspect of the environment and/or inert material such as bricks, concrete and other building debris, as well as tree roots and other organic matter *(Part Source: Contaminated Land Management Act 1997).*

Cut / cutting / ground excavation means lowering the natural ground surface by the regrading and/or the removal of soil from land.

Dam Dewatering Plan (DDP) is a document that sets out the measures that will be taken to reduce impacts on native wildlife when filling in or decommissioning dams or wetland areas. The plan also identifies and considers the risks of contaminated sediment and water disposal and protecting other water bodies from contamination and weed species.

Defined Flood Event (DFE) means the flood event selected as a general standard for the management of flooding to development. The 1% Annual Exceedance Probability flood has been selected as the Defined Flood Event in Blacktown.

Flood Planning Area (FPA) means the area of land below the Flood Planning Level.

Flood Planning Level (FPL) means the combination of the flood level from the Defined Flood Event and freeboard selected for flood risk management purposes.

Flood Prone Land means land susceptible to flooding by the PMF event. Flood prone land is also known as the floodplain, flood liable land and flood affected land.

Floodway means those areas of the floodplain where a significant discharge of water occurs during floods. They are often aligned with naturally defined channels. Floodways are areas that, even if only partially blocked, would cause a significant redistribution of flood flow, or a significant increase in flood levels.

Landfill means any work or activity which involves the placement of soil or other material on land, excluding the top dressing of lawns, whether undertaken as a principal activity or associated with another development, which in the opinion of Council significantly alters the shape or drainage of land and includes any earthworks or excavation which would have similar impact, whether or not such works would involve the importation of fill.

Natural ground surface / level means the natural existing land surface that has not been filled, cut, excavated, reshaped or otherwise altered.

Office of Environment and Heritage incorporates the NSW Environment Protection Authority.

Probable Maximum Flood (PMF) means the largest flood that could conceivably occur at a particular location, usually estimated from probable maximum precipitation (Probably

Maximum Precipitation), and where applicable, snow melt, coupled with the worst flood-producing catchment conditions.

Probable Maximum Precipitation (PMP) means the greatest depth of precipitation for a given duration meteorologically possible over a given size storm area at a particular location at a particular time of the year, with no allowance made for long-term climatic trends (World Meteorological Organization 1986).

Sediment means material of varying size, both mineral and organic, that is being, or has been, moved from its site of origin by the action of wind, water or gravity, and comes to rest on the earth's surface either above or below sea level.

Sedimentation means the deposition of sediment, usually in such locations as a channel, along a fence line or an area of low slope, or in a gully, creek, river, sediment trap or dam.

Soil means a natural material consisting of layers, amalgamates or individual particles of mineral and/or organic constituents, of variable thickness, that differs from its parent material in morphological, physical, chemical and mineralogical properties and biological characteristics.

Soil erosion means the detachment, entrainment, suspension and transport of soil material by wind, water or gravitational effects.

Unauthorised fill means the deposition of fill onto a site, which has not received approval from Council.

Virgin excavated natural material (VENM) means natural material such as clay, gravel, sand, soil and rock which is not mixed with any other type of waste and which has been excavated from areas of land that are not contaminated.

Waste, as defined under the *Protection of the Environment Operations Act 1997*, includes any:

(a) Substance (whether solid, liquid or gaseous) that is discharged, emitted or deposited in the environment in such volume, constituency or manner as to cause an alteration in the environment, or
(b) Discarded, rejected, unwanted, surplus or abandoned substance
(c) Any otherwise discarded, rejected, unwanted, surplus or abandoned substance intended for sale or for recycling, processing, recovery or purification by a separate operation from that which produced the substance, or
(d) Any processed, recycled, re-used or recovered substance produced wholly or partly from waste that is applied to land, or used as fuel, but only in the circumstances prescribed by the regulations, or
(e) Substance prescribed by the regulations to be waste.

A substance is not precluded from being waste for the purposes of the Act merely because it can be processed, recycled, re-used or recovered.

2 What is landfill?

Landfill involves the deposition of soil or other material on a site after it has been excavated/cut and transported from another location, whether on-site or off-site.

Some examples of landfill activities include:

(a) Reshaping an area of land, dam or floodway (involving cutting, grading, excavating or landfilling)

(b) Rural earthdams

(c) Retaining walls

(d) Building platforms.

Development proposals involving landfill generally require Council's consent. The only exception is where a proposal is considered to be a type of exempt development under the *State Environmental Planning Policy (Exempt and Complying Development Codes) 2008*.

A DA for landfilling could also constitute integrated development and/or designated development pursuant to Schedule 3 of the *Environmental Planning and Assessment Act 1979* and Regulations. Separate additional requirements will need to be met if these legislative requirements apply.

3 General requirements

A person who undertakes any landfill activity will have to demonstrate through the application process that they have considered the potential impacts on:

(a) Soil erosion and sedimentation

(b) Flooding / floodways / stormwater runoff

(c) Groundwater / surface water

(d) Contaminated material (existing or imported)

(e) Existing trees and vegetation

(f) Water pollution

(g) Dust (air pollution).

These requirements are explained in Sections 3.2 to 3.4 of this Part.

Landfilled areas must:

(a) Comprise clean material free from contamination (imported material shall be virgin excavated natural material (VENM)

(b) Be suitably compacted and stabilised with density tests to verify that compaction was achieved

(c) Be revegetated where appropriate

(d) Be provided with soil erosion and sedimentation controls in accordance with Part J (Water Sensitive Urban Design and Integrated Water Cycle Management) of Blacktown DCP 2015.

3.1 Relevant policies, legislation and guidelines

There are a range of Council policies, State Government legislation and guidelines that place obligations on any person who undertakes landfill-related activities. These are set out below.

(a) Blacktown City Council policies and planning instruments

The following local plans and policies may place an obligation on any person who undertakes any excavation / landfill activities, namely:

(i) Blacktown DCP 2015 Part A (Introduction and General Guidelines) Section 5 Biodiversity, Vegetation and Tree preservation and Section 10 Development on flood prone land

(ii) Blacktown Development Control Plan 2015 Part G (Site Waste Management and Minimisation)

(iii) Blacktown DCP 2015 Part I (Contaminated Land Guidelines)

(iv) Blacktown DCP 2015 Part J (Water Sensitive Urban Design and Integrated Water Cycle Management)

(v) Eastern Creek Floodplain Management Study and Plan 1996

(vi) South Creek Floodplain Management Study 1999.

(b) State legislation and guidelines

The following State legislation and guidelines may place an obligation on any person who undertakes any excavation / landfill activities, namely:

(i) State Environmental Planning Policy (Exempt and Complying Development Codes) 2008

(ii) Contaminated Land Management Act 1997

(iii) Environmental Planning and Assessment Act 1979 and Regulations 2000

(iv) Fisheries Management Act 1994

(v) Local Government Act 1993

(vi) National Parks and Wildlife Act 1974 No 80

(vii) Flood Risk Management Manual: The management of flood liable land (Department of Planning and Environment, June 2023)

(viii) Protection of the Environment Operations Act 1997

(ix) Rivers and Foreshores Improvement Act 1948

(x) Soil Conservation Act 1938

(xi) Sydney Regional Environmental Plan No. 20 - Hawkesbury-Nepean River (No 2 – 1997)

(xii) Biodiversity Conservation Act 2016

(xiii) Waste Avoidance and Resource Recovery Act 2001.

(c) Commonwealth legislation

The Environment Protection and Biodiversity and Conservation Act 1999 may place an obligation on any person who undertakes any excavation / landfill activities, which is deemed to be a Controlled Action under the Act.

3.2 Requirements for lodging a Development Application

There are specific planning considerations for proposed landfill activities when an applicant lodges a DA, as outlined below.

All work must comply with *Blacktown Development Control Plan 2015* Part A Section 5 Biodiversity, Vegetation and Tree preservation and Section 10 Development on flood prone land.

(a) Location of landfill

In general, landfilling within the flood planning area will not be

supported. However, Floodplain Landfill Applications will be treated on their merits based on the following:

(i) No net loss of flood storage and/or conveyance within the floodway extents

(ii) No net loss of flood storage within the 1% annual exceedance probability (AEP) critical duration flood. This requirement must address the cumulative impacts on flood levels from like development on other areas of flood prone land.

(iii) The alteration of local drainage or overland flow contours and/or natural watercourses must not adversely affect adjacent property.

Council may require the undertaking of detailed hydrologic / hydraulic modelling and survey contour plans in support of such applications.

Where landfilling is proposed within 40m of a watercourse, under the *Water Management Act 2000* a Controlled Activity Approval must be obtained from the NSW Department of Climate Change, Energy, the Environment and Water.

Where work involves the reconstruction of creek lines, reclamation or dredging works, a separate permit will be required from the Department of Primary Industries under the *Fisheries Management Act 1994.*

Where work impacts areas of land on the NSW Biodiversity Values Map, habitat for threatened species or will clear vegetation above the 'clearing area threshold' (see *Blacktown Development Control Plan 2015* Part A), a separate Biodiversity Development Assessment Report will need to accompany the proposal.

Where works involve the loss of farm dams or wetlands, whether natural or constructed; a Dam Dewatering Plan is required to accompany the proposal.

(b) Imported landfill

Where landfill is proposed to be imported onto land, a report must be submitted to Council which must be prepared by a NSW Office of Environment and Heritage accredited consultant. The report must verify that the landfill material is suitable for the intended use of the site having regard to the presence and levels of possible contaminants. The report is to be approved by Council prior to any earthworks commencing on the site.

(c) Slope of the site

In the instance where landfill occurs on a site where the natural slope exceeds 1 in 8 (12.5%), details of any proposed batters or retaining works shall be indicated. The proposed cross-fall for batters used in cut and landfill are to be restricted to a maximum slope of 1(V) in 4(H) (25% - the maximum slope allowable for general maintenance).

(d) Temporary stockpiles

Where the temporary stockpiling of landfill material will occur on any

land, this must be in accordance with Part J (Water Sensitive Urban Design and Integrated Water Cycle Management) of Blacktown DCP 2015.

(e) Archaeological sites

Any cut / landfill proposal can potentially damage or destroy existing Aboriginal artefacts on a site and consideration must be given to such sites.

Within Part A (Introduction and General Guidelines) of Blacktown DCP 2015 Council has established several different classes of archaeological sites. These are:

(i) Known archaeological sites

These sites which have been recorded by the NSW Office of Environment and Heritage are indicated on Council's DCP maps which are available for viewing online on Council's website at www.blacktown.nsw.gov.au.

(ii) Areas of high archaeological significance

These are areas with a high likelihood of archaeological sites occurring in these locations and are shown in detail on Council's DCP maps. The applicant is required to consult with the NSW Office of Environment and Heritage.

(iii) Areas of potential archaeological significance

Within these areas there is a possibility of sites occurring. The applicant will be required to consult with the Office of Environment and Heritage to determine if there are any requirements as a condition of consent.

(iv) Areas of low archaeological significance

Areas other than those noted above are substantially developed. However, where it becomes the case that a site of low archaeological significance is found, a Development Consent for work will require consultation with the Office of Environment and Heritage.

It is therefore paramount that, prior to any landfill or other work being determined, a relevant site be inspected by representatives of the relevant local Aboriginal groups to determine the archaeological significance of the land. The OEH is able to assist with current stakeholder contact information. This consultation will inform whether a detailed archaeological report may be required by a suitably qualified archaeologist.

The bulk earthworks phase may also require an inspection by the relevant Aboriginal group/s to determine if there are any artefacts that have been, or could potentially be, disturbed by earthworks.

(f) Flora and fauna

A Flora and Fauna Assessment report is required for all proposals that do not trigger a Biodiversity Development Assessment Report but do

potentially impact upon:

(i) Land mapped as Cumberland Plain Vegetation 2022; Conservation zoned land, Riparian corridors and protection Areas and/or Terrestrial Biodiversity in the Blacktown GIS Map layers or via the links provided at Part A Section 5.2

(ii) Native vegetation which is potential habitat for threatened species, populations or part of an endangered ecological community as identified in Section 1.6 of the *Biodiversity Conservation Act 2016* or the *Fisheries Management Act 1994*

(iii) Natural and constructed wetlands, dams or other wet areas. Note. A dam dewatering plan is required

(vi) Stream creek works in riparian channels e.g., bridges, basins.

The Flora and Fauna Assessment must include any five-part Test of Significance under Section 7.3 of the *NSW Biodiversity Conservation Act 2016* that may be required. Any likely significant impact will result in the requirement for further investigation and the preparation of either a Species Impact Assessment Statement or a Biodiversity Development Assessment Report.

The Flora and Fauna Assessment is to give regard to Clause 7.2 Terrestrial biodiversity and 7.3 Riparian land and watercourses of *Blacktown Local Environmental Plan 2015*.

(g) Other considerations

Should it be proposed to vary the finished ground or floor levels from those shown on the approved DA plans, then re-notification to neighbouring owners and residents may be required. Council may exercise its rights under Section 4.55 of the *Environmental Planning and Assessment Act 1979* and seek a formal request to modify the development consent.

Furthermore, the Environmental Impact Study Guidelines – Landfilling prepared by the NSW Department of Urban Affairs and Planning in 1996, is to be used when preparing an Environmental Impact Study. Consultation must also be undertaken with NSW Planning and Environment.

(h) Other consultations

Additional consultations may be required. The table below provides a list of the types of development that would require additional consultation:

Type of development	Authority
Works to prevent flooding, such as levees	NSW Department of Climate Change, Energy, the Environment and Water
Dam, weir or structures to impound water	NSW Department of Climate Change, Energy, the Environment and Water
Within 40m of a watercourse or which diverts a watercourse or affects the quantity of water flowing into or from a river or lake	NSW Department of Climate Change, Energy, the Environment and Water
Development involving dredging or reclamation in a waterway	NSW Department of Primary Industries NSW Department of Climate Change, Energy, the Environment and Water

Table 3.1 Type of development

3.3 Requirements for supporting documentation

A DA must be accompanied by the following supporting documents where applicable:

(a) A statement providing full justification for the proposed landfilling works. The statement should include the details of alternatives to the proposed landfilling option. This information is to be submitted with landfill / excavation activities that require Council consent.

(b) Suitable plans drawn to an appropriate scale (1:100, 1:200, 1:500, 1:1,000 or 1:2,000 should be used to adequately define detail) and indicating:

(i) Levels to be provided at Australian Height Datum (AHD)

(ii) Finished (to be shown in bold) and existing or natural ground level contours, to be displayed as follows:

Total existing ground relief over subject site:

- <1m – contours at 100mm
- <2m – contours at 200mm
- >2m – contours at 500mm.

Sufficient levels to accurately define contours must be taken. Particular care should be taken at changes in grade, natural watercourses / creek lines and other major features.

(c) The height above existing ground levels on adjoining properties is to be shown where retaining walls are used for retention of landfill on the site.

(d) Batter slopes of a maximum 5 (horizontal) to 1 (vertical) are to be shown for the tapering edges of landfilling works, where no retaining walls are intended.

(e) Tree Survey to identify vegetation species and their location.

(f) Traffic Report, detailing the transportation of landfill, may be required, dependent on the volume of landfill required to be transported to a site.

(g) Hydraulic / Hydrological Report by a chartered professional Civil Engineer or equivalent which addresses cumulative impacts (eg the displacement of flood storage and effect on flood levels in flood prone areas) may be required.

(h) Soil Contamination Report may be required to identify the source of the landfill. Such a report would need to be prepared by a NSW Office of Environment and Heritage accredited consultant.

3.4 Operational compliance

Where Council has granted consent to a DA, and all works have taken place, Council may request evidence of compliance. This could be in the form of:

(a) Works as Executed Plans
(b) Free from contamination verification / certification
(c) Surveyor's report
(d) Certification of hydraulic recommendation

from suitably qualified persons.